

When studying a species in nature, scientists often begin by asking, what environmental factors limit where it is found? How do variations in an organism's food supply or interactions with other species, such as predators, affect the size of their populations?

Questions like these are the subject of **ecology**, the scientific study of the interactions between organisms and the environment. (Note that here and throughout this text,

the term *environment* refers to other organisms as well as the physical aspects of an organism's surroundings.) The interactions studied by ecologists can be organized into a hierarchy that ranges in scale from single organisms to the planet (**Figure 52.2**). We'll begin our study of ecology by considering how Earth's climate and other factors determine the location of major life zones on land and in the oceans.

▼ Figure 52.2 Exploring the Scope of Ecological Research

Ecologists work at different levels of the biological hierarchy, from individual organisms to the planet. Here we present a sample research question for each level of the hierarchy.

Organismal Ecology

Organismal ecology, which includes the subdisciplines of physiological, evolutionary, and behavioral ecology, is concerned with how an organism's structure, physiology, and behavior meet the challenges posed by its environment.

◀ How do flamingos select a mate?

Population Ecology

A **population** is a group of individuals of the same species living in an area. **Population ecology** analyzes factors that affect population size and how and why it changes through time.

◀ What environmental factors affect the reproductive rate of flamingos?

Community Ecology

A **community** is a group of populations of different species in an area. **Community ecology** examines how interactions between species, such as predation and competition, affect community structure and organization.

◀ What factors influence the diversity of species that interact at this African lake?

Ecosystem Ecology

An **ecosystem** is the community of organisms in an area and the physical factors with which those organisms interact. **Ecosystem ecology** emphasizes energy flow and chemical cycling between organisms and the environment.

◀ What factors control photosynthetic productivity in this aquatic ecosystem?

Landscape Ecology

A **landscape** (or seascape) is a mosaic of connected ecosystems. Research in **landscape ecology** focuses on the factors controlling exchanges of energy, materials, and organisms across multiple ecosystems.

◀ To what extent do nutrients from terrestrial ecosystems affect organisms in the lake?

Global Ecology

The **biosphere** is the global ecosystem—the sum of all the planet's ecosystems and landscapes. **Global ecology** examines how the regional exchange of energy and materials influences the functioning and distribution of organisms across the biosphere.

◀ How do global patterns of air circulation affect the distribution of organisms?

